An interesting aspect of PCA is the connection of (1) with the quadratic form maximization formulations found in the descriptions of quantum problems. In particular, in VQE and given a Hamiltonian $M \in \mathbb{C}^{p\times p}$, the ground state energy, $E_{\min} \in \mathbb{R}$, is always upper-bounded by the expectation of M with respect to a trial wavefunction. That is, $E_{\min} \leq \langle \psi | M | \psi \rangle$ for any $\psi \in \mathbb{C}^p$. Similarly, the maximum energy, $E_{\max} \in \mathbb{R}$, is always lower bounded as in $E_{\max} \geq \langle \psi | M | \psi \rangle$ for any $\psi \in \mathbb{C}^p$.

What is different in VQE is the way we evolve the putative solution: Given an initial point $|\psi_0\rangle$, we start exploring from that point through the dynamics of the variational form $|\psi(\theta)\rangle = \prod_{i=0}^{\ell-1} U_i(\theta_i)|\psi_0\rangle$, where $U_i(\theta_i)$ is a user-defined/designed unitary matrix. In more detail, an ansatz is prepared through a set of parameters $\theta \in \mathbb{R}^m$ to calculate the eigenvector/*eigenstate* corresponding to the largest or least eigenvalue of M via:

$$\max_{\theta\in\mathbb{R}^m} / \min \quad \langle \psi(\theta) | M | \psi(\theta) \rangle \quad \text{s.t.} \quad |\psi(\theta)\rangle = \prod_{i=0}^{\ell-1} U_i(\theta_i)|\psi_0\rangle. \tag{2}$$

Here, the dimension $p = 2^q$ acts on q qubits; $|\psi(\theta)\rangle \in \mathbb{C}^p$ is a quantum state, parameterized by m variables $\theta \in \mathbb{R}^m$; $U_i(\theta_i) \in \mathbb{C}^{p\times p}$ represents a layer of an ansatz, repeated $\times\ell$ times. See Figure 1 for a depiction. Using the bra-ket notation above, the analogs of *i)* $v \leftrightarrow |\psi(\theta)\rangle$, *ii)* $x^\top y \leftrightarrow \langle x|y\rangle$, and *iii)* $\|v\|_2 = 1 \leftrightarrow \||\psi(\theta)\rangle\|_2 = 1$, are determined between PCA and VQE. The key differences are the fact that PCA operates directly on v with no restrictions other than forcing v to be normalized, while, in VQE, we are looking for a parameterized vector $|\psi(\theta)\rangle$ through a specific evolution from an initial state as in $|\psi(\theta)\rangle = \prod_{i=0}^{\ell-1} U_i(\theta_i)|\psi_0\rangle$. Details of the ingredients of (2) are in the text below.

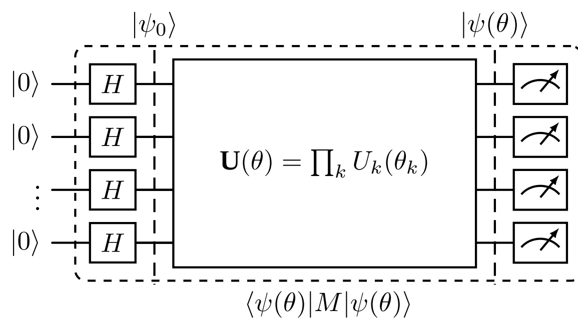

Figure 1: Circuit schematic for VQE where $|\psi(\theta)\rangle$ is prepared as $|\psi(\theta)\rangle = \mathbf{U}(\theta)|\psi_0\rangle$ when $|\psi_0\rangle = |+\rangle$. The initial layer of Hadamard gates may be excluded to have $|\psi_0\rangle = |0\rangle$. The circuit is then measured over an observable M to retrieve $\langle \psi(\theta) | M | \psi(\theta) \rangle$.

We note that solving the maximization or minimization problem in (1) *or* (2) *is an equivalent problem, since e.g.,* $\min\langle \psi(\theta) | M | \psi(\theta) \rangle = \max -\langle \psi(\theta) | M | \psi(\theta) \rangle$ *by applying a sign operator. For the rest of the text, we will focus on the maximization case.*

Algorithm 1 Ideal VQE with Deflation

Input: $M \in \mathbb{C}^{p\times p}$, `VQE` routine for ground state calculation, # of excited states k, and iters t.

$\Psi = \emptyset$
$M_1 \leftarrow M$
for $j = 1 : k$ **do**
  $|\psi_j(\theta)\rangle \leftarrow$ `VQE` (M_j, t) in (7)
  $\lambda_j \leftarrow \langle \psi_j(\theta) | M_j | \psi_j(\theta) \rangle$
  $M_{j+1} \leftarrow M_j - \lambda_j |\psi_j(\theta)\rangle\langle\psi_j(\theta)|$
  $\Psi \leftarrow \Psi \cup \{|\psi_j(\theta)\rangle\}$
end for
Return Set Ψ of estimated eigenstates.

Beyond "top" eigenstates. Multicomponent PCA looks for K principal components, which means finding the K most excited states in a VQE setting. For $M \in \mathbb{C}^{p\times p}$ with sorted true principal components $|\psi_1^\star\rangle, |\psi_2^\star\rangle, \dots, |\psi_p^\star\rangle$, we are interested in finding vectors $|\psi_1\rangle, \dots, |\psi_k\rangle$, where $k \leq p$, such that the difference between the corresponding true eigenvector $|\psi_i^\star\rangle$ and $|\psi_i\rangle$, for all $i \in [k]$, is bounded as in $\||\psi_i\rangle - |\psi_i^\star\rangle\|_2 \leq \varepsilon$, for some desired bound ε. One way to handle such a case is through deflation [37]: once the largest component $|\psi_1^\star\rangle$ is approximated, M is further processed to "live" on the subspace orthogonal to the subspace spanned by this component. The process then continues by again applying single-component VQE on the deflated M, which leads to an approximation of the second component (second excited state) $|\psi_2^\star\rangle$, and so on.

How could quantum deflation look in math? At the $(j+1)$-th iteration of deflation, we estimate the leading eigenvector of:

$$\max_{\theta\in\mathbb{R}^m} \quad \langle \psi(\theta) | M_{j+1} | \psi(\theta) \rangle \quad \text{s.t.} \quad |\psi(\theta)\rangle = \prod_{i=0}^{\ell-1} U_i(\theta_i)|\psi_0\rangle, \tag{3}$$